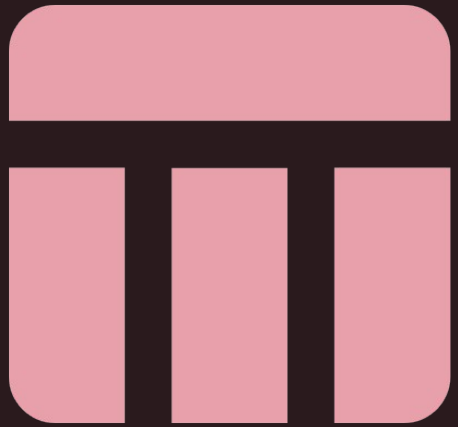
LAB - DAY 1

Higher National Diploma in Network Engineering, Year 2
BSc (Hons) Ethical Hacking & Network Security, Year 2

Mapping data, properties and risk



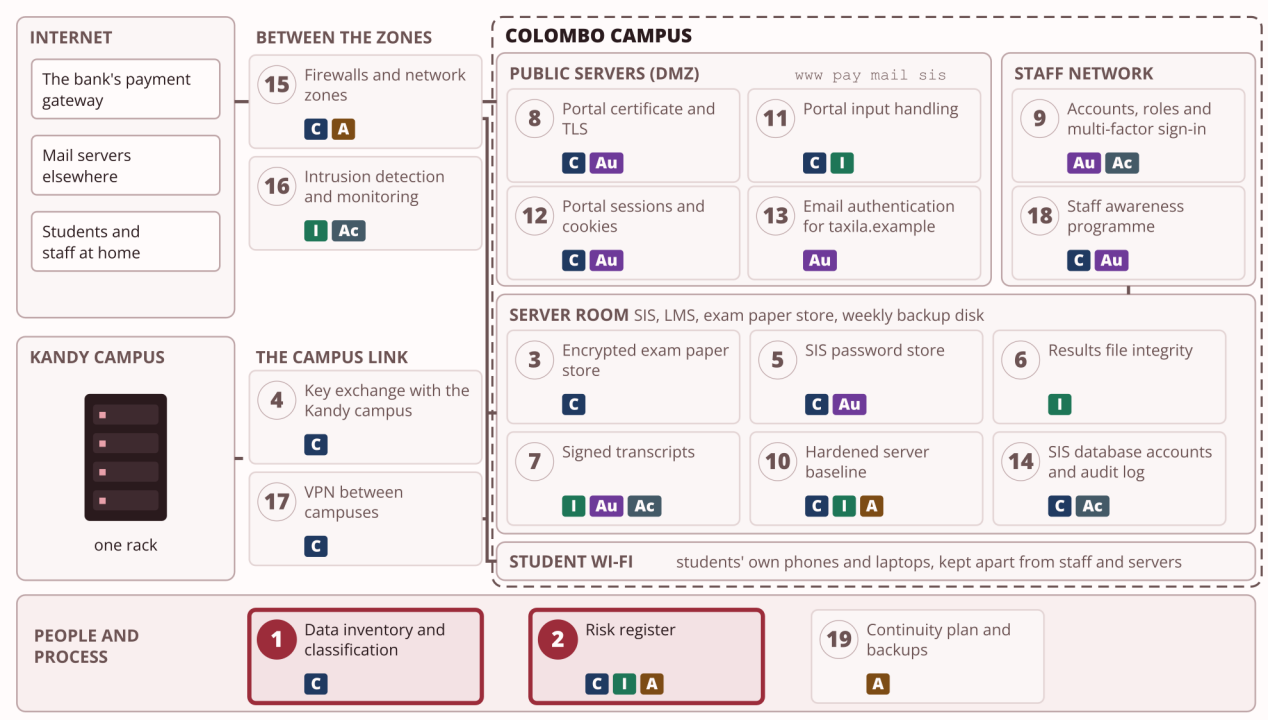
Adds to the register: 1 2

Floor tier: every step

Full tier: not needed

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What this lab adds



TODAY

1 Data inventory and classification

2 Risk register

KEY

Added today

Not yet added



DATA PROTECTION OFFICER

Whose personal data sits in each restricted and confidential row, and which three rows should we fix first?

The Data Protection Officer, at the start of the lab.

SEGMENT 1 - 45 MIN

Classify Taxila's data

Floor tier only

Classify Taxila's data

1 Open `data-inventory.csv`: seventeen rows, numbered 1 to 17.

Expected: Each row names a system and one kind of data.

2 Name each row's structure: structured, semi-structured or unstructured.

Expected: Student records are structured; mailboxes are unstructured.

3 Give each row a class: public, internal, confidential or restricted.

Expected: Exam papers and identity card numbers are restricted; the catalogue is public.

Floor path: The whole lab is floor tier: use the classification page in the notes.

One test for every row: how much harm would a leak cause? Not who owns the data.

The worksheet you fill

```
Row,System,Data kind,Structure,Class,Stage,Property,Why
2,Student Information System,Grades,,,,,
10,Exam paper store,Exam papers before the exam,,,,,
11,Website,Course catalogue,,,,,
```

SEGMENT 2 - 45 MIN

Map the lifecycle and the property at stake

Floor tier only

Where is each kind most exposed?

4 Walk grades (row 2) through create, store, process, share, archive, destroy.

*Expected: Share and store stand out; **integrity** is most at stake.*

5 Do the same for identity card numbers (row 4).

*Expected: Store and share again; this time **confidentiality**.*

6 Fill stage and property for every restricted and confidential row.

Expected: Confidentiality is common, but not the only answer.

Floor path: Same on every laptop; no tool needed.

SEGMENT 3 - 40 MIN

Write the first risk-register rows

Floor tier only

A risk row has five parts and a score

```
Asset:          Restricted exam papers before the exam
Threat:         A curious student
Vulnerability: A shared folder anyone at the college can open
Likelihood 3 x Impact 5 = 15
Property:       Confidentiality
```

The example row from the template. Read it before you write your own.

Write ten rows

7 Copy `risk-register.csv` and read the example row.

Expected: Asset, threat, vulnerability, 3, 5, 15, confidentiality.

8 Fill R01 to R10: one real Taxila asset each, scored 1 to 5 twice.

Expected: Each row names a real asset and one property; scores between 1 and 25.

Floor path: Same on every laptop.

9 Cover the threat cell, then the vulnerability cell: is it still a risk?

*Expected: Every row needs **both**; a weakness with no threat is not a risk yet.*

SEGMENT 4 - 20 MIN

Prove it, check it, and write up

Floor tier only

Prove it, check it, write up

10 Swap worksheets with a partner and find one row you classed differently.

Expected: Each of you can name the harm your class rests on.

11 Sort your ten rows by score, highest first.

Expected: The top rows are where the Head of IT looks first.

Prove it, then break it

PROVE IT

- Exam papers (row 10) and identity card numbers (row 4) are classed **restricted**
- One sentence each says why, in terms of harm

× BREAK IT

- Class grades as internal: a lecturer could now publish them
- Correct the class and name the property it protects

Reset, and the rules

RESET

- Nothing to reset; this lab changes no system
- Keep both files for Day 9 and your group

RULES OF ENGAGEMENT

- No tool in this lab can affect anyone else's system
- You work only with the provided worksheet and your own files

Before the next day

READ	**Security in Computing, chapter 1** Harm and the value of assets, with the classification page in the notes.
PRACTICE	**Practice question 1** A district clinic's appointment system. *Due before the next teaching day.*
TASK	**Bring your classification and risk rows** They grow into your group's register later.

Questions & Discussion

Next: Day 2, secrecy with keys.



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